

Subword Tokenizer and SLM Training/Evaluation Stack: A Practical Foundation for Physics Research Assistant Systems

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Abstract

This paper presents an implementation-focused foundation for building a domain-oriented small language model (SLM) system for physics research assistance. The system combines a production Rust Byte Pair Encoding (BPE) tokenizer with a Python-based training and evaluation pipeline designed for resource-constrained hardware (Apple Silicon/MPS). We report Phase 1–3 results: baseline TinyLM loss 0.0107, best LoRA loss 0.0060 (44% improvement), held-out test loss 0.00595 (45.45% gain vs baseline), and improved physics QA score (+0.125 absolute). We document architecture, operational scripts, reliability mechanisms (checkpointing and stream retry/backoff), and a phased roadmap toward retrieval-augmented generation (RAG) and tool-using agents.

1 Introduction

Small language models are increasingly relevant for cost-efficient, low-latency, and on-device AI workloads. Our objective is to establish a robust foundation layer that is reproducible on laptop-class hardware while remaining directly extensible to larger-scale training and deployment.

At a basic level, our methodology follows the now-standard autoregressive transformer formulation for next-token prediction [7], but adapts implementation choices to constrained hardware and practical research workflows. We use subword tokenization based on byte-pair encoding principles [8] to control vocabulary size and improve robustness for scientific terminology. We then apply lightweight adaptation ideas aligned with LoRA-style parameter-efficient fine-tuning [9] and evaluate pathways toward retrieval-augmented generation (RAG) [10]. In short, the paper is less about proposing a new model architecture and more about providing a reproducible systems recipe that converts modern LLM/SLM concepts into a reliable, end-to-end physics-assistant stack.

The current system has two aligned layers:

1. **Tokenizer layer (Rust):** a production BPE tokenizer with CLI and reusable library APIs.
2. **Prototype training layer (Python):** a TinyLM training path using the Rust tokenizer model, with automatic checkpointing and evaluation.

2 System Architecture

The implementation follows a streaming-to-training architecture:

1. Live arXiv stream with retry/backoff for transient failures.
2. Tokenization through Rust CLI (`bpe-tokenizer`) using a 32K vocabulary model.



Figure 1: System architecture: Live arXiv stream → Rust tokenizer → TinyLM training → checkpoint evaluation.

3. Tiny Transformer training (MPS/CUDA/CPU auto-selection).
4. Step checkpoints, final checkpoints, and multi-run ranking by evaluation loss.

3 Implementation Details

3.1 Tokenizer Commands

Current tokenizer workflow:

- `cargo run -bin bpe-tokenizer - train <corpus.txt> <vocab_size>`
- `cargo run -bin bpe-tokenizer - expand <corpus.txt> <vocab_size>`
- `cargo run -bin bpe-tokenizer - tokenize <text>`
- `cargo run -bin bpe-tokenizer - decode <comma_separated_ids>`
- `cargo run -bin bpe-tokenizer - count`

3.2 Prototype Training Commands

- Quick prototype: `./scripts/run_prototype.sh`
- 3-run compare + best select: `./scripts/run_prototype_3x_and_select_best.sh`
- Long run (~3–4h on M3 Pro): `./scripts/run_prototype_long_4h.sh`
- Evaluation: `./scripts/run_evaluation.sh`

3.3 Validation and Reliability

- Rust tokenizer tests are passing.
- Prototype training is validated on Mac MPS.
- Best-checkpoint JSON reports are generated automatically.
- Retry/backoff mitigates arXiv timeout interruptions in long runs.

4 Roadmap to Publication-Grade Physics Assistant SLM

4.1 Final Goal

Build a physics research assistant using a 3B–7B parameter SLM, LoRA fine-tuning, RAG grounding, and tool-using agents.

4.2 Phase Plan

Weeks	Phase	Deliverable
1–2	Data Pipeline	300M–1B tokenized physics corpus
3–5	Base SLM Training	3B–7B checkpoint (decoder-only baseline)
6–7	LoRA Fine-Tuning	Parameter-efficient domain adaptation weights
8–9	RAG Integration	Vector index + retrieval-augmented response path
10–11	Agent Layer	Tool-using multi-step reasoning workflow
12+	Deployment	API + optional UI + cloud/runtime packaging

4.3 Success Criteria

- Physics-grounded coherent answers with retrieved context.
- Stable tool-calling behavior for equations and code tasks.
- End-to-end response latency below 5 seconds in target deployment profile.

5 Why This Work Is Useful for the Community

This work is intended to be useful for three overlapping communities. First, for students and independent researchers, it demonstrates a realistic path to building and validating domain-specialized language systems on commodity hardware, reducing dependence on large cluster budgets. Second, for applied ML engineers, it provides operational patterns (checkpointing, retry/backoff, evaluation scripts, and reproducible artifacts) that are often under-documented in purely model-centric papers. Third, for domain communities (physics and adjacent sciences), it offers a transparent scaffold for building assistants that can be audited, extended, and grounded via retrieval, rather than treated as opaque black-box systems.

More broadly, the project contributes implementation clarity: which pieces are essential, what metrics matter at each phase, and where practical bottlenecks appear (e.g., early-EOS generation behavior despite improved loss metrics). This type of report can help others avoid costly iteration cycles and establish comparable baselines for future open-domain and domain-specific SLM research.

6 Risks and Mitigations

- **Streaming instability:** addressed via retry/backoff and periodic checkpoints.
- **Artifact growth:** requires retention and storage policy.
- **Documentation drift:** enforce CI-based PDF generation from source.

7 Conclusion

This work establishes a practical and reproducible foundation for SLM development, balancing implementation realism with a clear path to research-grade publication outcomes. The current milestone is a validated tokenizer+training+evaluation stack (Phase 1–3 complete); the next milestones are generation-control tuning, RAG integration, and agent-layer deployment.

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